

Post-Flight Evaluation of Stardust Sample Return Capsule Forebody Heatshield Material

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Phenolic Impregnated Carbon Ablator (PICA) was developed at NASA Ames Research Center under the lightweight ceramic ablator development program in the '80s. PICA's low density ($\sim 0.27\text{g/cc}$), coupled with efficient ablative capability at high heat fluxes, makes it an enabling technology for the Stardust mission. This paper discusses the evaluation of three cores extracted post flight from key locations on the forebody heatshield of the Stardust Sample Return Capsule (SRC). Core locations included a near stagnation core, a flank core, and a segment taken from the shoulder of the heatshield. Evaluation included density profiles, recession determination, a thermal analysis profile, PICA bondline examination, strength assessment of remaining virgin PICA, an emissivity profile, a chemical analysis profile, and a microstructural analysis. Results show good agreement in comparisons of experimental density profiles and profiles derived from FIAT and in recession comparisons from measured values and FIAT predictions for the flank core. In general, the PICA material examined in the cores is in good condition and intact. Impact damage is not evident, and the only degradation observed was that caused by heating on entry. A substantial amount of virgin PICA was present in all cores examined.

Nomenclature

m_{PICA} = mass loss (PICA)
 $m_{Phenolic}$ = mass loss (phenolic)

I. Introduction

Stardust was the initial U.S. space program to attempt sample return from beyond the orbit of the Moon. The mission performed a flyby of the comet Wild-2, deploying a sample collection apparatus to gather extraterrestrial material from the comet coma. The spacecraft, launched in February 1999, returned to Earth in January 2006 and released a sample return capsule (SRC), which entered the earth's atmosphere and was recovered. The entry velocity ($\sim 12.9\text{km/s}$) and heating ($\sim 1100\text{W/cm}^2$ at the stagnation) of the Stardust SRC were the highest of any Earth entry to date.¹

PICA, a new thermal protection system (TPS), was developed for the forebody heatshield of the SRC because conventional TPS materials of the time (primarily carbon phenolics) had high densities and thermal conductivities, yielding a TPS mass fraction that exceeded mission constraints. PICA was developed at NASA Ames Research Center under the lightweight ceramic ablator development program in the '80s and consists of a rigid carbon fibrous substrate infiltrated with phenolic resin, yielding a TPS with good ablation and pyrolysis behavior.² In addition, PICA has the advantages of low density ($\sim 0.27\text{g/cc}$) coupled with efficient ablative capability at high heat fluxes that made PICA an enabling technology for the Stardust mission.

The Stardust heatshield was 0.78 meters in diameter and was manufactured as a single piece, to promote fiber orientation parallel to the vehicle surface, thereby minimizing the through-the-thickness thermal conductivity. It is

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worth noting that, at the time of the Stardust mission, PICA was, in essence, a developmental material, with no previous flight heritage. The PICA process was transferred from NASA Ames to Fiber Materials, Inc. (FMI) of Biddeford, Maine, and the PICA heatshield was manufactured at the FMI facility. The PICA heatshield was directly bonded to the substructure (an aluminum honeycomb with graphite/polymer facesheets), using a commercially available high-temperature adhesive (HT-424, a Cytotec product).

II. Core Location, Extraction, and Segmentation

Cores at three key locations were extracted from the forebody heatshield, post flight, and evaluated. Core locations include a near stagnation core, a flank core, and a PICA segment taken from the exterior edge of the heatshield, as indicated in the core location map in Figure 1.

Cores were extracted with a diamond-tipped core drill for the flank and near stagnation cores, and a straight blade for the edge piece. The cylindrical cores were two inches (5.08cm) in diameter. The near stagnation and flank core were examined destructively, while the edge piece has received minimal evaluation at this time and remains intact.

Cores (two inches in diameter) were cleaned after extraction to eliminate remaining curf, which covered the external surfaces, and to reduce potential surface contamination. In general, the cores were in excellent condition, with substantial virgin PICA remaining in all cores, as shown in Figure 2. The near stagnation and flank cores were then further segmented, and half of each core was used for this study. The remaining half of each core is retained in curation at Johnson Space Center.



Figure 1. Forebody Core Location Map indicating position of near stagnation, flank cores and shoulder piece.

III. Material Characterization

A number of techniques were employed to primarily evaluate the near stagnation and flank cores. Minimal evaluation was also completed on the edge piece. Evaluation included density profiles, recession determination, a thermal analysis profile, PICA bondline examination, strength assessment of remaining virgin PICA, an emissivity profile, a chemical analysis profile, and a microstructural analysis. The details of each technique and results will be summarized in this section.

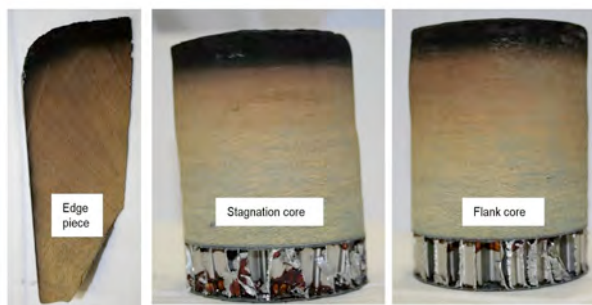


Figure 2 As extracted cores after cleaning

A. Density Profile of Near Stagnation and Flank Cores

The density profile, as a function of location from the char surface, was obtained for both cores by sectioning the cores in submillimeter segments, determining the density of the segment from its mass and volume, and documenting the location of the segment from the char surface.

Experimental error in density measurements is estimated at ± 4 percent, whereas error in location from the char surface is estimated at ± 2 percent. The density profile of the near stagnation and flank cores is plotted in Figure 3. This figure also includes a series of images of each segment used to determine the density profile for the stagnation core.

As evident from Figure 3, it is possible to visually estimate the transition from char to pyrolysis zone to virgin material for PICA systems. Visual observations on transitions in PICA from segments agree with the density profile generated for both cores. For example, in the near stagnation core, we observed densities approaching the virgin material at slice 10, which corresponds with an observed color transition approaching virgin in this slice.

B. Recession Determination

Recession at both the near stagnation and flank cores is estimated knowing both the pre- and post-flight TPS thicknesses. We determined recession at the near stagnation location to be 0.57 cm (0.22 inches) and recession at the flank location to be 0.32 cm (0.126 inches). Estimated error in measured recession is ± 3 percent. The error is based on measurement limitations and on the fact that the cores were not extracted perfectly normal to the substructure surface.

C. Thermal Analysis (thermogravimetric) Profile

A series of thermogravimetric analysis (TGA) runs helped to better understand the decomposition behavior of PICA as a function of location from the char surface. TGA runs corresponding to the core slices used for density profiles were used for both the near stagnation and flank cores. All runs were in argon with a heating rate of 10° Celsius per minute. From the TGA profiles, we can estimate remaining phenolic resin (unpyrolyzed) as a function of location from the char surface using the following relationship:

$$\% \text{ Resin} = \frac{m_{PICA}}{m_{phenolic}} \times 100 \quad (1)$$

where m_{PICA} is the mass loss in the PICA and $m_{phenolic}$ is the mass loss in phenolic after pyrolysis (estimated at 37.4 percent from reference 2). The resin content, as a function of location for both the near stagnation and flank cores, is shown in Figure 4. It should be noted that there is some error in these measurements, since the sample size analyzed was smaller than optimal in all cases. As expected, the estimated

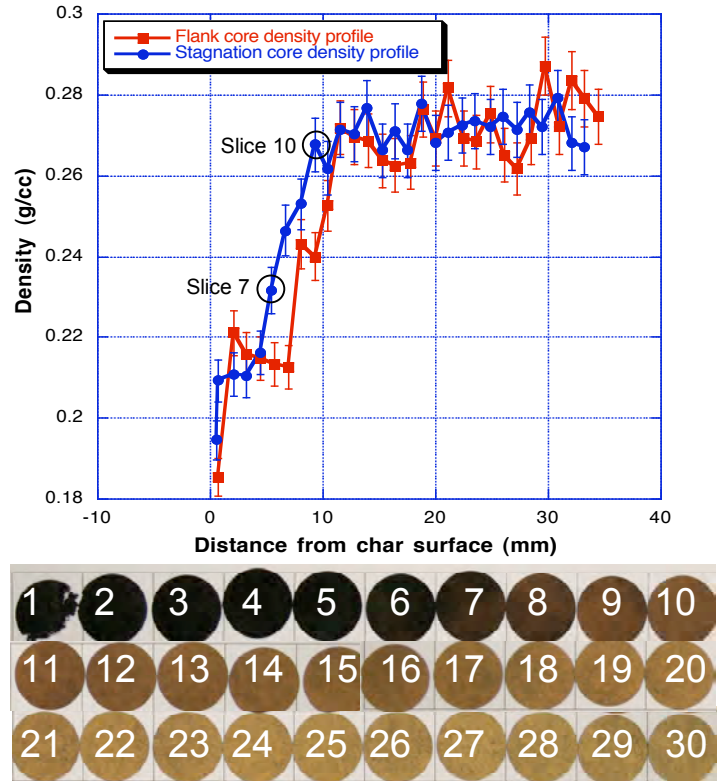


Figure 3. Density profile of near stagnation and flank cores along with images of segments used to obtain profile for near

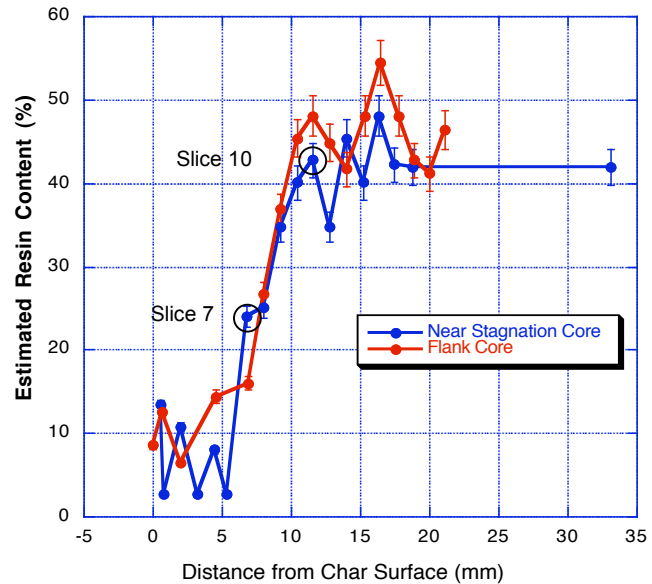


Figure 4. Estimated resin content as a function of location from the char surface

resin content as a function of location trace follows a similar trend to the density profile obtained for each core in Figure 3. Taking the example of the near stagnation core in Figure 4, the transition to the pyrolysis zone starts at approximately 5mm from the char surface. A comparison to the density profile of the near stagnation core in Figure 3 shows the same transition to initiate at approximately 5mm from the char surface.

D. Bondline Examination

We examined the PICA-composite substructure bondline, using microscopy, to investigate debonding or adhesion issues. We found no evidence of questionable bond integrity for either core. A typical micrograph of the PICA-composite substructure bondline is shown in Figure 5.

E. Strength of Remaining Virgin PICA

Sufficient virgin PICA remained in each core to evaluate the strength of the material and determine if some degradation had occurred in mechanical performance as a result of exposure to the space environment or as a result of landing impact. We employed a simple pull test to determine strength. Samples consisted of cylinders one inch (2.54cm) in diameter with heights of greater than 0.3 inches (0.76cm). From in-house analysis, it was determined that a minimum of 0.3 inches (0.76cm) of virgin PICA was required for uniform axial stress distribution. Unfortunately, only one sample could be obtained from each core, so the data set is very limited. Through-the-thickness (TTT) strength values of 43.8 psi (302 kPa) and 39.1 psi (269.6kPa) were obtained for the near stagnation and flank cores, respectively. Heritage testing of virgin PICA yielded an average TTT strength of 39.4 psi (271.7kPa), which is in the range observed for both cores. This preliminary testing indicates that exposure to the space environment did not mechanically degrade the virgin material.

F. Microstructural Profiles

Variation in PICA morphology, as a function of depth, was also evaluated. Figure 6 presents micrographs at known locations in the near stagnation core, including a typical virgin PICA microstructure, where the two constituents (carbon fibers and a high surface area phenolic phase) are easily identified. Note that the morphology of charred PICA at the surface (the initial 1 to 2 mm of surface char) does not contain the high surface area phase present at all other locations imaged. This supports the density observations in Figure 3, where a density of less than 0.2g/cc is observed close to the char exterior surface. Commercially available fiberform (the substrate from which PICA is processed) commonly has densities approaching 0.18g/cc. Degradation of the fibers present in the char was not significant, as the fiber diameter of the char is comparable to the fiber diameter of the baseline carbon fiber substrate material used to make PICA, indicating minimal damage to the surface fibers.

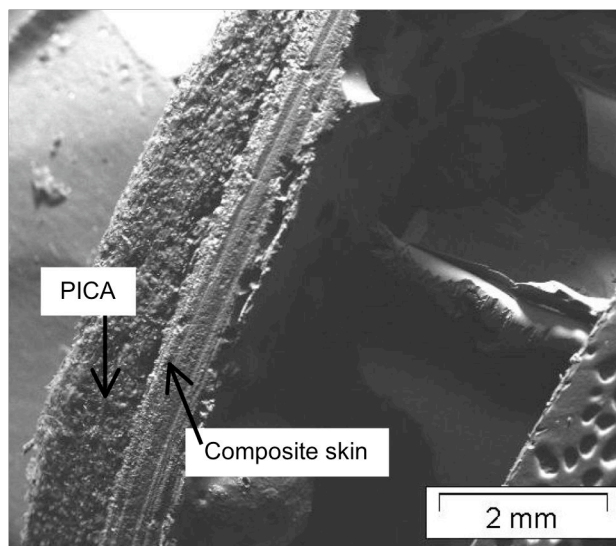


Figure 5. Representative micrograph of the PICA-composite substructure bondline

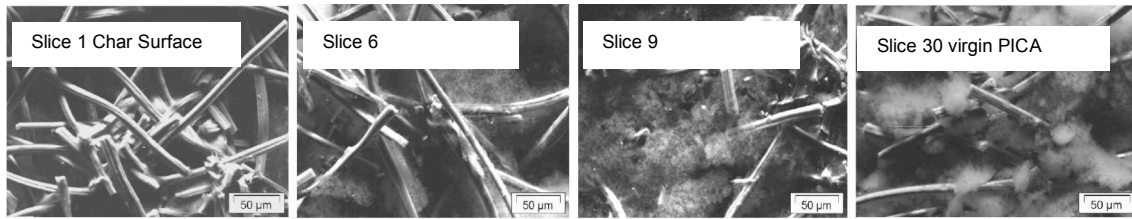


Figure 6. Representative morphologies at various locations along the near stagnation core

G. Emissivity Profiles

We measured the spectral reflectance of each core, as a function of distance from the char surface, at room temperature over a wavelength range of 0.25 mm to 2.5 mm. We determined the emissivity of individual core slices (corresponding to the core slices used for density profiles) for both the near stagnation and flank cores and data for the stagnation core is given in Figure 7. While the emissivity as a function of location trace follows a similar trend to the density profile obtained for each core in Figure 3 and the resin content trace Figure 4, we observed some differences. For example, a transition from char to pyrolysis zone is evident at slice 6 (as observed in resin content and density traces); however, after slice 6, a gradual transition to virgin material occurs, over many millimeters of sample, and virgin is not observed until slices 18 through 19. In comparison, the density and resin content trace indicate a more rapid transition to virgin occurring at approximately slice 10. The gradual transition in the emissivity trace indicates that alteration occurs in the phenolic due to a thermal exposure; however the exposure is not at sufficiently high temperatures or duration to result in phenolic decomposition.

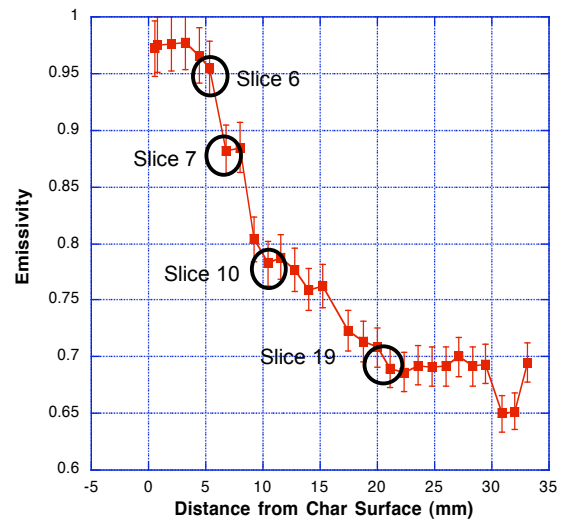


Figure 7. Emissivity Profile of Near Stagnation Core as a function of distance from char surface

H. Other Ongoing Analysis

We have obtained chemical analysis profiles along the core depth (for both cores) and are comparing these data to the information obtained from the airborne observation, to help explain some of the chemical traces observed during entry. Elements of interest include sodium, potassium, calcium, and zinc. Some of these elements are probably contaminants present in virgin PICA or introduced as a result of the thermal control coating. A more complete study of the chemical analysis data is ongoing. In an in-progress collaborative effort with Stanford University, microprobe laser-desorption laser-ionization mass spectrometry analysis is being used to compare stardust flight heritage PICA to PICA tested in arc jet facilities. This work is supporting ground-to-flight traceability. In addition, NASA Goddard is developing a coupled UV and spectroscopic technique to determine the heating history experienced by the Stardust cores. If successful, this would be very useful data, because Stardust was not instrumented, and the heating history of the TPS is not available to support model development.

IV. Comparing Experimental Data with Predicted Data

The experimentally obtained density profile and recession were compared using measured data from the heatshield and calculated data based on best-known trajectories. The predicted values were derived from FIAT,³ (Fully Implicit Ablation and Thermal response code), developed at NASA Ames Research Center in the 1990s for modeling TPS (Thermal Protection System) response to atmospheric entry. FIAT solves internal energy balance and decomposition equations, coupled with the ablating surface energy balance condition, to simulate the response of

ablative heatshields in hypersonic flows, and has been used to perform TPS sizing for many of NASA's planetary missions. FIAT is a reliable and stable numerical procedure for use under a wide range of surface and in-depth conditions. In addition to removing the potential numerical instability problem of the CMA code, FIAT code has additional features, such as an integrodifferential equation for internal radiative flux, optimized TPS thickness, and a flow code interface.

I. Recession

Using pre- and post-flight TPS thicknesses, the recession at the near stagnation and flank cores is measured (section IIIA) with an error of ± 5 percent. The corresponding recession values were calculated using FIAT v2.4⁴ and the PICA v3.3 property model,⁵ considering convective and radiative heating. The resulting values are summarized in Table 1. For the near stagnation core, the model over predicts recession by 61 percent. For the flank core, the difference in the predicted recession values is 25 percent. The discrepancies at the flank core are of the same order as differences between calculation and arc-jet tests against which the model was calibrated.⁵ The over-prediction at the near-stagnation core is not fully understood. Sensitivity analysis is underway to identify the environmental and material model parameters that within their range of uncertainty lead to lower recession estimates.

Table 1. Measured and calculated recessions for near stagnation (C1) and flank (C2) cores (cm).

Location	Measured (cm)	Convective heating only		Convective and radiative heating	
		Calculated (cm)	% difference from measured	Calculated (cm)	% difference from measured
Stagnation Point	-	0.96	-	1.04	-
Core 1	0.57 $\pm$ 0.03	0.86	51	0.92	61
Core 2	0.32 $\pm$ 0.02	0.39	22	0.4	25

J. Density Profile

Figure 8 plots the measured and calculated density profiles. The same trends are evident in both measured and calculated profiles, where a larger char region is present in the flank core, as expected. There is relatively good agreement between the measured and predicted density profiles, indicating the response model captures the decomposition/ablation phenomenon.

V. Conclusion

PICA TPS was an enabling technology for the Stardust mission. In an effort to evaluate its performance, three cores were extracted from the forebody heatshield of the Stardust SRC. Core locations include a near stagnation core, a flank core, and a segment taken from the shoulder of the heatshield. Evaluation included density profiles, recession determination, a thermal analysis profile, PICA bondline examination, strength assessment of remaining virgin PICA, an emissivity profile, chemical analysis profiles, and microstructural analysis. Comparisons between experimental density profiles and profiles derived from FIAT are in good agreement. Recession comparisons from measured values and FIAT predictions are in good agreement for the flank core. In general, the PICA material examined in the cores is in good condition and intact. The recovered material shows comparable mechanical performance (in the virgin portion) to heritage strength data. Impact damage is not evident, and the only degradation observed was that caused by heating on entry. A substantial amount of virgin PICA was present in all cores examined.

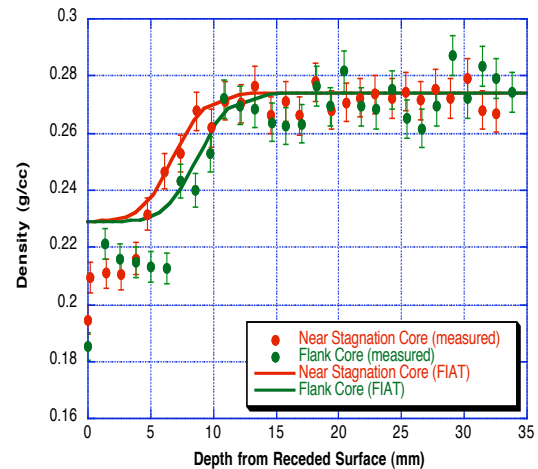


Figure 8. Comparison of measured and calculated density profiles for the near stagnation and flank cores. The x-axis is referenced to each receded surface.

Acknowledgments

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